

Remove Deprecated Volatile Features from C++26

Proposal to remove easily misunderstood feature

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1 Abstract

C++ has deprecated a number of features related to `volatile` semantics in both the core language specification and in the library specification. This paper proposes removing those features from C++26.

2 Revision History

R4: July 2024 (post-St Louis mailing)

- Recorded Core working group review from St Louis
- Update Core wording plan to explain subtle changes
 - moved wording plan to the C++26 Review section
 - described why the current plan no longer touches 12.5 [\[over.built\]](#)
- Sent to SG22 to capture any concerns regarding C compatibility
- Raised EWG concerns for removal of volatile-qualified function parameters
- Updated core wording
 - fixed typos and grammar nits highlight in Core review
 - restored support for `operator=` as unevaluated operand
 - cleaned up the wording for the pre-increment operators
 - fixed the specification for compound assignment to a volatile lvalue
 - removed changes to 12.5 [\[over.built\]](#) as unnecessary
 - simplified rationale for removing support for `operator=`
 - updated Annex C examples to use preferred ISO form
- Updated library wording
 - describe `tuple_element` and `tuple_size` as type traits rather than metafunctions
 - completed the list of library types affected by removing volatile from tuple traits
 - unmerged the rationale for removing volatile variant support from volatile tuple support
 - drive-by fix to include specializations of tuple traits for const-qualified types in `<complex>`

R3: June 2024 (St Louis meeting)

- Recorded summary of the reviews for C++26 in all working groups
- Deferred non-deprecated changes to class template `atomic` to a new paper
- Updated wording
 - Rebased wording onto latest working draft, [\[N4981\]](#)
 - Provided missing rationales for Annex C

R2: April 2024 (post-Tokyo mailing)

- Updated wording
 - Rebased wording onto latest working draft, N4971
 - Annex C: changed “will not compile” to “may become ill-formed”
 - Simplified note that removing trait specializations for volatile types does not remove support for volatile-qualified elements

R1: September 2023 (midterm mailing)

- Removed revision history’s redundant subsection numbering
- Noted proposal passed EWG review, but awaiting LEWG confirmation before passing to Core
- Added SG22 C Interoperability to target audience
- Provided the missing Library Analysis
- Analyzed the remaining structured binding dependency on `volatile` in the library
- Updated wording
 - Applied initial Core wordsmith preview from Jens Maurer

- Rebased onto latest working draft, N4958
- Updated stable label cross-reference to C++23

R0: May 2023 (pre-Varna mailing)

Original version of this document, extracted from the C++23 proposal [[P2139R2](#)].

Key changes since that earlier paper

- Combined core and library updates in a single paper
- C++23 undeprecated compound assignment
- Rebased wording onto N4944

3 Introduction

At the start of the C++23 cycle, [P2139R2] tried to review each deprecated feature of C++ to see which we would benefit from actively removing and which might now be better undepricated. Consolidating all this analysis into one place was intended to ease the (L)EWG review process but in return gave the author so much feedback that the next revision of the paper was not completed.

For the C++26 cycle, a much shorter paper, [P2863R1], will track the overall analysis, but for features that the author wants to actively progress, a distinct paper will decouple progress from the larger paper so that the delays on a single feature do not hold up progress on all.

This paper takes up the deprecated operations on volatile types, D.4 [depr.volatile.type], and the associated deprecated library features.

4 Background

The `volatile` keyword is an original part of the C legacy for C++ and describes constraints on programs intended to model hardware changing values beyond the program's control. As this entered the type system of C++, certain interactions were discovered to be troublesome, and latent bugs that could be detected at the time of program translation go unreported. [P1152R4] breaks down each context where the `volatile` keyword can be used and it deprecated for C++20 those uses that are unconditionally dangerous or serve no good purpose.

Following the C++20 deprecations, the C committee looked to adopt a similar stance on `volatile` and were given feedback that a number of vendors were strongly opposed to the deprecation of compound-assignment operators because, among other reasons, many hardware APIs and device drivers would expect to use volatile compound assignment to communicate with their devices. This subset of the deprecated functionality was undepricated for C++23 by [P2327R1], followed by further undeprecations in [CWG2654].

5 Feature Analysis

5.1 Core language

A quick micro-analysis suggests the main concerns of the first two paragraphs are read/modify/write operations where, by the nature of volatile objects, the value being rewritten may have changed since read and modified. This kind of pattern is most likely in old (pre-C++11) code using `volatile` as a poor proxy for atomic. Since we will have well over a decade of real atomic support in the language when C++26 ships, further encouraging such code (when compiled in the latest dialect) to adapt to the memory model and its stronger guarantees could be desirable.

The third paragraph addresses function arguments and return values. These temporary or elided objects are created entirely by the compiler and guaranteed to not display the uncertainty of value implied by the `volatile` keyword. As such, any use is redundant and misleading, so removing this facility sooner rather than later would be helpful and would also mean one fewer oddity to teach when learning (and understanding) the language. The biggest concern would be compatibility with C code that may still use this feature in its headers. To mitigate, we may consider removing volatile function parameters and return values for only functions with `extern "C++"` linkage.

The fourth paragraph considers the `volatile` qualifier in structured bindings, and can affect only code written since C++17, and when C++23 is published such use of the qualifier will have been deprecated as long as it was nondeprecated. It would be good to remove this now, before more deprecated code is written.

The recommendation is to remove support for these deprecated use cases from C++26 since we have had seven years of deprecation warnings as well as the usages' potential to diagnose hard-to-reproduce latent bugs for users to fix. Reviewing each of the four noted usages separately would also be possible, as would be removing only those features having the lowest risk from removal — notably paragraphs 3 and 4.

5.2 Library

Three distinct feature sets were deprecated as part of the deprecating `volatile` work for C++20. Both `std::tuple` and `std::variant` have an API to query how many elements or alternates a type contains, and another to query what the type of a given element or alternate is. These APIs support volatile-qualified `tuple` and `variant` types, yet a corresponding `get` API to retrieve the value of that type has never been available, making these, making these interfaces largely redundant.

The other use of `volatile` in the Standard Library is as part of the atomic APIs, and several overloads for `volatile` non-lock-free `std::atomic` that should be constrained to not exist, per the primary library specification, remain deprecated in Annex D.

5.2.1 Deployment experience

By testing the following program with all of the latest compilers and Standard Library implementations available through Godbolt Compiler Explorer, we discovered that none of the existing library implementations are warning on use of the deprecated `tuple` and `variant` APIs. Deeper analysis may be needed to confirm whether this is a library issue, or whether such usage is something the compiler finds difficult to warn about when using the `[[deprecated]]` attribute.

```
#include <tuple>
#include <type_traits>
#include <variant>

using TypeT = std::tuple<int, char, float> volatile;
using TypeV = std::variant<int, char, float> volatile;

static_assert(std::is_same_v<std::tuple_element<0, TypeT>::type, int volatile>);
static_assert(std::is_same_v<std::tuple_element_t<0, TypeT>, int volatile>);

static_assert(std::is_same_v<std::variant_alternative<0, TypeV>::type, int volatile>);
static_assert(std::is_same_v<std::variant_alternative_t<0, TypeV>, int volatile>);
```

6 C++23 Feedback

6.1 Initial EWG review

The following feedback was provided when this core language feature was originally discussed in the EWG telecon on July 30, 2020.

This clause is effectively four different sub-features, that were reviewed and polled independently. The author offered to pull this whole section out into another paper if there were concerns about processing a complex topic in this simplified omnibus paper (which has effectively happened in this paper), but relatively little contention arose throughout the discussion, so it will remain here for now.

Some concerns were raised that by removing some of these features, we would be creating inconsistencies between the treatment of `const` and `volatile` in the language. Others suggested that this inconsistency was a good thing and that one of the early concerns Bjarne expressed about the design and evolution of C++ was too much consistency in the treatment of these two qualifiers that do different things in practice.

The observation was made — several times — that `volatile` qualifiers on locally scoped variables, such as function arguments, rarely mean what naive users expect them to mean, and can be freely ignored by an optimizing compiler. By removing support for some of those declarations, we make writing misleading (but otherwise correct) code more difficult.

6.2 Subsequent feedback

Following feedback from WG14 and their progress for C23, reading the deprecated result of compound assignment to a `volatile` lvalue for the bitwise operators was undeprecated for C compatibility in C++23 by [P2327R1]. Subsequently, responding to NB comment US 16-045, reading the result of the remaining compound assignment operators was undeprecated by [CWG2654], reintroducing a potential C incompatibility in favor of consistency and a simpler language.

7 Proposed Changes for C++26

This paper proposes removing from C++26 all the deprecated features regarding the use of `volatile`.

7.1 Core language

Remove the following language interactions:

- increment and decrement operators on volatile lvalues
- `volatile` qualifier on non-reference function parameters
- `volatile` qualifier on non-reference function return types
- volatile-qualified structured bindings

In addition, built-in assignment operator functions for `volatile` lvalues should be declared to return `void`. C++23 deprecates calling assignment operators with volatile lvalues unless they are a discarded-value expression or an unevaluated operand. We can enforce this by simply removing the return value from the function signature. However, this change is bigger than strictly necessary since it further removes the nondeprecated use case as an unevaluated operand. This is the recommended choice as it means that code written to detect valid return types using SFINAE constraints will report only valid code; otherwise, we would risk breaking metaprograms.

7.2 Library

7.2.1 tuple API

Remove deprecated `tuple` traits of volatile-qualified types. I tried and failed to demonstrate the need to support a customization point of structured bindings of `volatile`-qualified types. Structured bindings of `volatile`-qualified `std::tuple` objects already fail to compile due to a lack of `get` support, and my test cases of trying to set up a user customization for their own types compiled without the `volatile` specializations.

7.2.2 variant API

Remove deprecated `variant` interface.

7.2.3 Non-lock-free atomics

Remove deprecated `volatile` members of `atomic<T>` when `atomic<T>::is_lock_free` is `false`.

7.2.4 (Deferred) Change volatile atomic interface to match nonatomic types

The following changes are deferred and *not* part of the proposed wording of this paper; instead, they shall be revisited in a separate, specific paper. The design questions to address are concerns that these functions have not yet been deprecated, unless their implementation is not lock-free. The forms that are not lock-free *are* removed by this paper, so we will not touch the specification for the forms that remain.

`atomic<integral-type>` and `atomic<pointer-type>` should remove `volatile`-qualified increment and decrement operators in all cases.

All nondeleted `volatile`-qualified `atomic<T>` assignment operators should change their return value to `void`, although this might be an ABI-breaking change.

In each case, the atomic operation returns a non-atomic value, rather than a reference to the `atomic` object itself, and so avoids the concerns that to removing these operations from the volatile scalars. However, it does seem odd to have operations on the volatile atomic types that do not exist on the corresponding volatile scalar types. That consistency is a design question better raised with SG1 though.

7.3 Concerns raised by core/library interaction

One corner case retains a core language dependency on `tuple_size` and `tuple_element` through structured bindings. Let us build up an example to demonstrate the specific corner that this paper proposes removing (without deprecation) from the Core specification, allowing LWG to remove the partial specializations of `tuple_size` and `tuple_element` for volatile-qualified types.

7.3.1 Tailored structured binding

First, demonstrate the feature that has been deprecated for the last two editions of the Standard. Below, we create a type in namespace `test`, `struct Binding`, that is a simple aggregate-like class that defines all the customization points necessary to use that type in a structured binding.

```
#include <tuple>
#include <type_traits>
#include <utility>

namespace test {
struct Binding {
    int data{};
    char code{};
    float value{};

    Binding() = default;
    Binding(Binding const&) = default;
};

template<unsigned N>
auto get(Binding& obj) {
    if constexpr (0 == N) {
        return obj.data;
    }

    if constexpr (1 == N) {
        return obj.code;
    }

    if constexpr (2 == N) {
        return obj.value;
    }

    std::unreachable();
}

template<unsigned N>
auto get(Binding const & obj) {
    if constexpr (0 == N) {
        return obj.data;
    }

    if constexpr (1 == N) {
        return obj.code;
    }

    if constexpr (2 == N) {
```

```

        return obj.value;
    }

    std::unreachable();
}

}

namespace std {

template <>
struct tuple_size<test::Binding> : std::integral_constant<unsigned, 3> {};

template <>
struct tuple_element<0, test::Binding> {
    using type = int;
};

template <>
struct tuple_element<1, test::Binding> {
    using type = char;
};

template <>
struct tuple_element<2, test::Binding> {
    using type = double;
};

}

int main() {
    test::Binding x = {};
    auto [a,b,c] = x;
    return a;
}

```

Here, explicitly specialize the templates `std::tuple_size` and `std::tuple_element` for our class type, and add `get` overloads in its own namespace that are found via ADL. This is a basic demonstration of supporting our own type in a structured binding.

7.3.2 Deprecated volatile structured binding

Then we update the main program:

```

int main() {
    test::Binding x = {};
    auto volatile [a,b,c] = x;
    return a;
}

```

This program gives warnings that this use of `volatile` is deprecated and is the usage this paper proposes removing.

7.3.3 Nondeprecated structured binding to a volatile-qualified lvalue

Next, we bind from a volatile-qualified lvalue instead:

```
int main() {
    test::Binding volatile x = {};
    auto [a,b,c] = x;
    return a;
}
```

This fails to compile as the structure binding wants to make a copy of `x`, but no constructor that can take a volatile-qualified argument is found, so we update `Binding` as follows:

```
struct Binding {
    int    data{};
    char   code{};
    float  value{};

    Binding() = default;
    Binding(Binding const&) = default;
    Binding(Binding const volatile &) {} // construct with default intializers
};
```

By overloading with the `const volatile &` copy constructor, the program with the volatile-qualified `x` now compiles:

```
int main() {
    test::Binding volatile x = {};
    auto [a,b,c] = x;    // this will compile now
    return a;
}
```

Note that this use of `volatile` is not deprecated and should remain supported. However, we may be wondering if this uses `tuple_size` on a volatile-qualified type? So let us test that!

Define explicit specializations of `tuple_size` for all cv-qualified variations of `test::Binding` so that only the unqualified version provides the integral constant base characteristics required by the structured binding protocol:

```
template <>
struct tuple_size<test::Binding> : std::integral_constant<unsigned, 3> {};

template <>
struct tuple_size<test::Binding const> {};           // canary

template <>
struct tuple_size<test::Binding volatile> {};        // canary

template <>
struct tuple_size<test::Binding const volatile> {};  // canary
```

If the structured binding attempted to find the `tuple_size` of a volatile-qualified object, it should fail to compile; however, our program continues to compile just fine, indicating that structured bindings are querying the non-volatile-qualified copy of `x` used for the by-value binding. Hence, this valid use of `volatile` is not impacted by the proposal to remove `volatile` support from `tuple_size` (and `tuple_element`).

7.3.4 Binding by reference to a volatile-qualified lvalue

Now, let us try to make a structured binding by-reference to a volatile lvalue. Note that according to the core language wording this is well-defined behavior that is not deprecated in C++23:

```
int main() {
    test::Binding volatile x = {};
    auto & [a,b,c] = x;
    return a;
}
```

Here we find that the structured binding relies upon the deprecated library value `tuple_size<volatile test::Binding>::val` suggesting that our proposal would break this code. However, retaining the volatile-qualified support for `tuple_size` is not yet enough for the above code to compile, even in the original C++17 specification that preceded the deprecations. The remaining issue is that our `get` overloads do not accept references to volatile-qualified types. Hence, to complete our implementation and as required for C++17, which is not affected by any changes proposed by this paper, we must add the ADL-discoverable volatile-qualified overloads:

```
template<unsigned N>
auto get(Binding volatile & obj) {
    if constexpr (0 == N) {
        return obj.data;
    }

    if constexpr (1 == N) {
        return obj.code;
    }

    if constexpr (2 == N) {
        return obj.value;
    }

    std::unreachable();
}

template<unsigned N>
auto get(Binding const volatile & obj) {
    if constexpr (0 == N) {
        return obj.data;
    }

    if constexpr (1 == N) {
        return obj.code;
    }

    if constexpr (2 == N) {
        return obj.value;
    }

    std::unreachable();
}
```

Adding these two overloads is sufficient to create a structured binding by-reference to our volatile-qualified lvalue. Note that the `get` overloads in namespace `std` for native arrays, `std::array`, `std::pair`, and `std::tuple` do not support volatile-qualified objects and never have. Hence, support for reference bindings to volatile lvalues has only ever been supported for user-provided types that supply the necessary ADL-discoverable overloads of `get`.

If we adopt this proposal to remove support for volatile-qualified types in the `tuple` metafunctions, then users will have to add their own specializations for `tuple_size` and `tuple_element` for their own type in addition to

their existing `get` overloads:

```
template <>
struct tuple_size<test::Binding> : std::integral_constant<unsigned, 3> {};

template <>
struct tuple_size<test::Binding const>
    : tuple_size<test::Binding>::type {};

template <>
struct tuple_size<test::Binding volatile>
    : tuple_size<test::Binding>::type {};

template <>
struct tuple_size<test::Binding const volatile>
    : tuple_size<test::Binding>::type {};
```

Note that whether users are currently permitted to specialize `tuple_size` in this way is unclear. However, if such specializations are not allowed, the original example and all like it are also not allowed, so no breakage of well-defined code would occur under this proposal.

7.3.5 Proposed resolution

We pre-emptively reject any proposal that would, without a period of deprecation, disqualify structured binding to volatile-qualified lvalues. Perhaps such a deprecation was intended by the original paper, [P1152R4], but if so, that has not been explicitly drafted.

The only breakage that occurs by removing the `tuple_size` and `tuple_element` specializations is binding by-reference to a volatile-qualified lvalue, and that already requires a user to provide partial and/or explicit specializations of the primary template for their type as well as a larger set of `get` overloads in their namespace (or as template-member functions of the class) than provided in namespace `std` for standard types, supporting volatile-qualified objects.

Assuming a user has already done all of the above so that their program would fail to compile with C++26, the specification is already clear on how users can fix their programs: The compiler is going to look for those specializations of `tuple_size` and `tuple_element` for their type, and being a user-provided type, the users can provide those specializations themselves.

Hence, the recommendation of this paper is to remove the deprecated `tuple` API, and maybe add a note to the structured bindings clause to suggesting how the user may support this edge case, in addition to any Annex C wording.

8 C++26 Design Review

8.1 EWG initial review: Varna 2023

There were some concerns with removing the support for top-level volatile qualifiers in structured binding, but a stronger consensus to proceed with the full proposal as written.

Forward to Core after the paper passes LEWG review

8.2 SG1 initial review: Varna 2023

Reviewed the section of this paper that has specific concerns for SG1, notably the deprecated `atomic` members that should SFINAE away when `is_lock_free` is `false`.

Consensus to forward to LEWG for final review, despite concerns about removing any feature forcing breakage on working code.

8.3 LEWG initial review: Kona 2023

Author plays devil’s advocate to ensure that concerns are heard, but the room is comfortable that the less the library says about volatile the better, so it would be good to remove these last few vestiges that are not needed.

Forward the library component of “P2866R1 Remove Deprecated Volatile Features From C++26” (8.2, and parts of 8.3) to LWG, to be confirmed by electronic polling.

The paper was submitted to the December 2023 LEWG electronic poll, [P3053R0], and was forwarded to LWG following a successful result [P3054R0].

9 C++26 Wording Review

9.1 Wording plan for core clauses

First, where we want to restrict operations to modifiable lvalues that no longer support `volatile`-qualified types, we will call out “modifiable non-volatile lvalues”, which excludes all *cv*-qualifiers, so we can strike *cv*-qualification too.

When it comes to using the result of a volatile-qualified assignment expression we note that by removing the deprecated support from the grammar, we never get into the situation that a volatile return types in 12.5 [over.built] are encountered as part of a valid expression. Rather than change the grammar to return `void` in such cases, we prefer to make the minimal change and leave this part of the grammar untouched, minimizing the risk of accidental breakage should we discover some path into these calls in an unevaluated operand where the changing the return type would be observable.

Finally, we remove support for the `volatile` qualifier without a reference qualifier when declaring a structured binding.

9.2 LWG initial review: Tokyo 2024

Several corrections were made live during review and are incorporated in the proposed wording below. Most significant changes were simplifying the note that removing trait specializations for volatile types does not remove support for volatile-qualified elements, as the original phrasing was quite confused, and broadly updating Annex C wording in all related papers to use the phrase “may become ill-formed” rather than “will not compile”.

9.3 Core initial review: St Louis 2024

Other than simple typos and grammar nits, concerns were raised about making the assignment operator ill-formed for unevaluated operands, as it should be well-formed to have an unevaluated operand that would be a discarded-value expression; a wording fix was suggested that preserved all the intended properties.

Concerns were raised about removing support for volatile qualified function parameters, as at least one attendee believed they had an active use case. A paper that instead proposed undeprecating volatile function parameters will be written, and sent to EWG along with this paper to confirm the preferred direction at the next meeting. No such concerns were raised about the function return types.

Concerns were raised about the changes to signatures in 12.5 [\[over.built\]](#), with the belief that the changes in 7 [\[expr\]](#) already pre-empt hitting the problem cases, and it would be safer to make only the minimal necessary change.

Further concerns were raised whether SG22, the C/C++ compatibility study group, had seen this paper. The next revision will be sent to SG22 for feedback.

10 Proposed Wording Changes

Make the following changes to the C++ Working Draft. All wording is relative to [N4981], the latest draft at the time of writing.

10.1 Update core wording

10.1.1 Changes to core clauses

7.6.1.6 [expr.post.incr] Increment and decrement

- ¹ The value of a postfix ++ expression is the value of its operand.

[Note 1: The value obtained is a copy of the original value. —end note]

The operand shall be a modifiable non-volatile lvalue. The type of the operand shall be an arithmetic type other than ~~cv~~ bool, or shall be a pointer to a complete object type. ~~An operand with volatile-qualified type is deprecated; see D.4 [depr.volatile.type].~~ The value of the operand object is modified (3.1 [defs.access]) as if it were the operand of the prefix ++ operator (7.6.2.3 [expr.pre.incr]). The value computation of the ++ expression is sequenced before the modification of the operand object. With respect to an indeterminately-sequenced function call, the operation of postfix ++ is a single evaluation.

[Note 2: Therefore, a function call cannot intervene between the lvalue-to-rvalue conversion and the side effect associated with any single postfix ++ operator. —end note]

The result is a prvalue. The type of the result is the ~~cv-unqualified version of the~~ type of the operand.

- ² The operand of postfix -- is decremented analogously to the postfix ++ operator.

[Note 3: For prefix increment and decrement, see 7.6.2.3 [expr.pre.incr]. —end note]

7.6.2.3 [expr.pre.incr] Increment and decrement

- ¹ The operand of prefix ++ or -- shall ~~not be of type cv bool. An operand with volatile-qualified type is deprecated; see D.4 [depr.volatile.type].~~ be a modifiable non-volatile lvalue. The type of the operand shall be an arithmetic type other than bool, or shall be a pointer to a complete object type. The expression ++x is otherwise equivalent to x+=1 and the expression --x is otherwise equivalent to x-=1 (7.6.19 [expr.ass]).

[Note 1: For postfix increment and decrement, see 7.6.1.6 [expr.post.incr]. —end note]

7.6.19 [expr.ass] Assignment and compound assignment operators

- ¹ The assignment operator (=) and the compound assignment operators all group right-to-left. All require a modifiable lvalue as their left operand; if the left operand is volatile qualified and the operator is the simple assignment operator, the program is ill-formed unless the (possibly parenthesized) assignment is a discarded-value expression or an unevaluated operand (7.2.3 [expr.context]), in which case the result is a prvalue of type void; otherwise their result is an lvalue of the type of the left operand, referring to the left operand. The result in all cases is a bit-field if the left operand is a non-volatile bit-field. In all cases, the assignment is sequenced after the value computation of the right and left operands, and before the value computation of the assignment expression. The right operand is sequenced before the left operand. With respect to an indeterminately-sequenced function call, the operation of a compound assignment is a single evaluation.

...

- ⁵ ~~An assignment whose left operand is of a volatile-qualified type is deprecated (D.4 [depr.volatile.type]) unless the (possibly parenthesized) assignment is a discarded-value expression or an unevaluated operand (7.2.3 [expr.context]).~~
- ⁶ The behavior of an expression of the form E1 op= E2 is equivalent to E1 = E1 op E2 except that E1 is evaluated only once, and if E1 is volatile qualified the result is an lvalue referring to E1.

9.3.4.6 [dcl.fct] Functions

2 ...

- 3 The *parameter-declaration-clause* determines the arguments that can be specified, and their processing, when the function is called.

[Note 1: The *parameter-declaration-clause* is used to convert the arguments specified on the function call; see 7.6.1.3 [expr.call]. —end note]

If the *parameter-declaration-clause* is empty, the function takes no arguments. A parameter list consisting of a single unnamed parameter of non-dependent type `void` is equivalent to an empty parameter list. Except for this special case, a parameter shall not have type *cv void*. A parameter ~~with~~ shall not have a volatile-qualified type ~~is deprecated; see D.4 [depr.volatile.type]~~. If the *parameter-declaration-clause* terminates with an ellipsis or a function parameter pack (13.7.4 [temp.variadic]), the number of arguments shall be equal to or greater than the number of parameters that do not have a default argument and are not function parameter packs. Where syntactically correct and where “...” is not part of an *abstract-declarator*, “, ...” is synonymous with “...”.

[Example 1: The declaration

```
int printf(const char*, ...);
```

declares a function that can be called with varying numbers and types of arguments.

```
printf("hello world");
printf("a=%d b=%d", a, b);
```

However, the first argument must be of a type that can be converted to a `const char*`. —end example]

[Note 2: The standard header `<stdarg.h>` (17.13.2 [stdarg.syn]) contains a mechanism for accessing arguments passed using the ellipsis (see 7.6.1.3 [expr.call] and 17.13 [support.runtime]). —end note]

- 4 The type of a function is determined using the following rules. The type of each parameter (including function parameter packs) is determined from its own *parameter-declaration* (9.3 [dcl.decl]). After determining the type of each parameter, any parameter of type “array of T” or of function type T is adjusted to be “pointer to T”. After producing the list of parameter types, any top-level ~~cv-qualifiers~~ const-qualifier modifying a parameter type ~~are~~ is deleted when forming the function type. The resulting list of transformed parameter types and the presence or absence of the ellipsis or a function parameter pack is the function’s *parameter-type-list*.

5 ...

- 14 The return type shall be a non-volatile non-array object type, a reference type, or ~~cv~~ possibly const-qualified `void`.

[Note 8: An array of placeholder type is considered an array type. —end note]

- 15 A volatile-qualified return type is deprecated; see 13.7.4 [temp.variadic].

9.6 [dcl.struct.bind] Structured binding declarations

- 1 A structured binding declaration introduces the *identifiers* `v0`, `v1`, `v2`,... of the *attributed-identifier-list* as names of *structured bindings*. The optional *attribute-specifier-seq* of an *attributed-identifier* appertains to the structured binding so introduced. Let *cv* denote the *cv-qualifiers* in the *decl-specifier-seq* and *S* consist of the *storage-class-specifiers* of the *decl-specifier-seq* (if any). Let *cv* denote the *cv-qualifiers* in the *decl-specifier-seq* and *S* consist of the *storage-class-specifiers* of the *decl-specifier-seq* (if any). A *cv* that includes `volatile` is ~~deprecated; see D.4 [depr.volatile.type]~~ ill-formed. First, a variable with a unique name *e* is introduced. If the *assignment-expression* in the *initializer* has array type *cv1 A* and no *ref-qualifier* is present, *e* is defined by

*attribute-specifier-seq*_{opt} *S* *cv* *A* *e* ;

and each element is copy-initialized or direct-initialized from the corresponding element of the *assignment-expression* as specified by the form of the *initializer*. Otherwise, *e* is defined as-if by

attribute-specifier-seq_{opt} decl-specifier-seq ref-qualifier_{opt} e initializer ;

where the declaration is never interpreted as a function declaration and the parts of the declaration other than the *declarator-id* are taken from the corresponding structured binding declaration. The type of the *id-expression* *e* is called **E**.

[*Note 1*: **E** is never a reference type (7.2 [expr.prop]). —end note]

² If the *initializer* refers to ...

⁵ Otherwise, ...

[*Example 2*:

```
struct S { mutable int x1 : 2; volatile double y1; };
S f();
const    auto [ x, y ] = f();
volatile auto [ a, b ] = f(); //error, volatile structured binding
```

The type of the *id-expression* **x** is “int”, the type of the *id-expression* **y** is “const volatile double”. —end example]

10.1.2 Update Annex C for differences to C++23

C.1.2 [diff.cpp23.expr] Clause 7: expressions

² **Affected subclause:** 7.6.1.6 [expr.post.incr] and 7.6.2.3 [expr.pre.incr]

Change: Cannot increment or decrement volatile scalars.

Rationale: The load and the store are not one atomic operation, which is a common source of bugs. Such operations should be clearly spelled out as **a = a + 1** to ensure the separate load and separate store are seen.

Effect on original feature: A valid C++ 2023 program that calls the increment or decrement operators on a volatile-qualified scalar object becomes ill-formed.

[*Example 1*:

```
volatile int x = 0;
++x; // ill-formed, OK in C++23
x--; // ill-formed, OK in C++23
```

—end example]

³ **Affected subclause:** 7.6.19 [expr.ass]

Change: Cannot use the return value of assignment to a volatile-qualified type.

Rationale: Relying on the return value of assigning to a volatile object is a subtle source of bugs that are not easy to observe by reading the code. Given the expression.

```
a = b = c;
```

where **a**, **b**, and **c** are lvalues of the same type, but **b** is also volatile, it is not immediately obvious that the assignment to **a** involves an additional load from **b**, which might produce a different value than **c**.

Effect on original feature: A valid C++ 2023 program that uses the result of assignment to a volatile-qualified object becomes ill-formed.

[*Example 2*:

```
volatile int b = 0;
int        c = b = 1; // ill-formed, in C++23 would assign 1 to b
               // and the subsequent value of b to c
```

—end example]

C.1.3 [diff.cpp23.dcl.dcl] Clause 9: declarations

² Affected subclause: 9.3.4.6 [dcl.fct]

Change: Cannot declare volatile-qualified function parameter types and function return types.

Rationale: Cv-qualification on parameter types apply only within the scope of the function definition, where volatile qualification is effectively meaningless as the compiler can see the whole lifetime of the object where there is no opportunity for some well-defined external process to change its value asynchronously.

Effect on original feature: A valid C++ 2023 program that declares volatile-qualified function parameters or functions with a volatile-qualified return type becomes ill-formed.

[Example 1:

```
int f(volatile int v) { return v; } // ill-formed, valid in C++23
```

—end example]

⁴ Affected subclause: 9.6 [dcl.struct.bind]

Change: Cannot define a volatile-qualified structured binding.

Rationale: Volatile qualified structured bindings require customization by users to support volatile qualified get calls, which is not possible for the templates in namespace `std` such as `pair` and `tuple`.

Effect on original feature: A valid C++ 2023 program that declares a volatile-qualified structured binding becomes ill-formed.

[Example 2:

```
int main() {
    int x[] = {1, 2, 3};
    std::tuple y{1, 2, 3};

    auto volatile [a, b, c] = x; // ill-formed, valid in C++23
    auto volatile [d, e, f] = y; // always ill-formed
}
```

—end example]

10.1.3 Update Annex C for differences to ISO C

C.7.4 [diff.expr] Clause 7: expressions

^x Affected subclause: 7.6.1.6 [expr.post.incr] and 7.6.2.3 [expr.pre.incr]

Change: Cannot increment or decrement volatile scalars.

Rationale: The load and the store are not one atomic operation, which is a common source of bugs. Such operations should be clearly spelled out as `a = a + 1` to ensure the separate load and separate store are seen.

Effect on original feature: Deletion of semantically well-defined feature.

Difficulty of converting: Semantic transformation.

How widely used: Seldom.

^y Affected subclause: 7.6.19 [expr.ass]

Change: Cannot use the return value of assignment to a volatile-qualified type.

Rationale: Relying on the return value of assigning to a volatile object is a subtle source of bugs that are not easy to observe by reading the code. Given the expression

```
a = b = c;
```

where `a`, `b`, and `c` are lvalues of the same type, but `b` is also volatile, it is not immediately obvious that the assignment to `a` involves an additional load from `b`, which might produce a different value than `c`.

Effect on original feature: Deletion of semantically well-defined feature.

Difficulty of converting: Semantic transformation.

How widely used: Seldom.

C.7.6 [diff.dcl] Clause 9: declarations

× **Affected subclause:** 9.3.4.6 [dcl.fct]

Change: Cannot declare volatile-qualified function parameter types and function return types.

Rationale: Cv-qualification on parameter types apply only within the scope of the function definition, where volatile qualification is effectively meaningless as the compiler can see the whole lifetime of the object where there is no opportunity for some well-defined external process to change its value asynchronously.

Effect on original feature: Deletion of semantically well-defined feature.

Difficulty of converting: Semantic transformation.

How widely used: Seldom.

10.1.4 Strike core wording from Annex D

D.4 [depr.volatile.type] Deprecated volatile types

- ¹ Postfix `++` and `--` expressions (7.6.1.6 [expr.post.incr]) and prefix `++` and `--` expressions (7.6.2.3 [expr.pre.incr]) of volatile-qualified arithmetic and pointer types are deprecated.

[Example 1:

```
volatile int velociraptor;  
++velociraptor; // deprecated
```

—end example]

- ² Certain assignments where the left operand is a volatile-qualified non-class type are deprecated; see 7.6.19 [expr.ass].

[Example 2:

```
int neck, tail;  
volatile int brachiosaur;  
brachiosaur = neck;           // OK  
tail = brachiosaur;          // OK  
tail = brachiosaur = neck;    // deprecated  
brachiosaur += neck;          // OK
```

—end example]

- ³ A function type (9.3.4.6 [dcl.fct]) with a parameter with volatile-qualified type or with a volatile-qualified return type is deprecated.

[Example 3:

```
volatile struct amber jurassic();           // deprecated
void trex(volatile short left_arm, volatile short right_arm); // deprecated
void fly(volatile struct pterosaur* pteranodon); // OK
```

—end example]

- ⁴ A structured binding (9.6 [\[dcl.struct.bind\]](#)) of a volatile-qualified type is deprecated.

[Example 4:

```
struct linhenykus { short forelimb; };
void park(linhenykus alvarezsauroid) {
    volatile auto [what_is_this] = alvarezsauroid; // deprecated
    // ...
}
```

—end example]

10.2 Update library wording

10.2.1 No changes to zombie names

As all the entities being struck are overloads of identifiers that retain their original meaning, no new names need be added to 16.4.5.3.2 [zombie.names].

22.4.7 [tuple.helper] Tuple helper classes

```
template<class T> struct tuple_size<const T>;
```

4 ...

6 In addition to being available via inclusion of the `<tuple>` header, the template is available when any of the headers `<array>` (24.3.2 [array.syn]), `<complex>` (28.4.2 [complex.syn]), `<ranges>` (26.2 [ranges.syn]), or `<utility>` (22.2.1 [utility.syn]) are included.

```
template<size_t I, class T> struct tuple_element<I, const T>;
```

7 ...

8 In addition to being available via inclusion of the `<tuple>` header, the template is available when any of the headers `<array>` (24.3.2 [array.syn]), `<complex>` (28.4.2 [complex.syn]), `<ranges>` (26.2 [ranges.syn]), or `<utility>` (22.2.1 [utility.syn]) are included.

10.2.2 Add Annex C Library wording for differences to C++23

C.1.X Annex D: compatibility features [diff.cpp23.depr]

✗ **Change:** Remove volatile support for volatile-qualified `array`, `complex`, `pair`, `ranges::subrange`, and `tuple` in the type traits `tuple_element` and `tuple_size`.

Rationale: The library does not make extra effort to support volatile types and the support offered by just these type traits without support from the function `get` provided little value.

Effect on original feature: A valid C++ 2023 program using these type traits for volatile-qualified `array`, `complex`, `pair`, `ranges::subrange`, or `tuple` may become ill-formed.

[Note N: This change does not remove support for volatile-qualified types stored in an `array`, `pair`, `ranges::subrange`, or `tuple`. —end note]

✓ **Change:** Remove volatile support for volatile-qualified `variant` in the type traits `variant_alternative`, and `variant_size`.

Rationale: The library does not make extra effort to support volatile types and the support offered by just these type traits without support from the function `get` provided little value.

Effect on original feature: A valid C++ 2023 program using these type traits for volatile-qualified `variant` may become ill-formed.

[Note N: This change does not remove support for volatile-qualified types stored in a `variant`. —end note]

✗ **Change:** Remove support for operations on volatile `atomic<T>` unless `atomic<T>::is_always_lock_free` is `true`.

Rationale: Implementations that are not able to be implemented lock-free risk introducing word tearing, which is not permitted for correct behavior of atomic operations.

Effect on original feature: A valid C++ 2023 program using such operations on an `atomic<T>` object may become ill-formed.

10.2.3 Strike Library wording from Annex D

D.15 [depr.tuple] Tuple

- ¹ The header (22.4.2 [tuple.syn]) has the following additions:

```
namespace std {  
    template<class T> struct tuple_size<volatile T>;  
    template<class T> struct tuple_size<const volatile T>;  
    template<size_t I, class T> struct tuple_element<I, volatile T>;  
    template<size_t I, class T> struct tuple_element<I, const volatile T>;  
}
```

```
template<class T> struct tuple_size<volatile T>;  
template<class T> struct tuple_size<const volatile T>;
```

- ² Let *TS* denote `tuple_size<T>` of the cv-unqualified type *T*. If the expression `TS::value` is well-formed when treated as an unevaluated operand (7.2.3 [expr.context]), then specializations of each of the two templates meet the *Cpp17TransformationTrait* requirements with a base characteristic of `integral_constant<size_t, TS::value>`. Otherwise, they have no member value.
- ³ Access checking is performed as if in a context unrelated to *TS* and *T*. Only the validity of the immediate context of the expression is considered.
- ⁴ In addition to being available via inclusion of the `<tuple>` (22.4.2 [tuple.syn]) header, the two templates are available when any of the headers `<array>` (24.3.2 [array.syn]), `<ranges>` (ranges.syn), or `<utility>` (22.2.1 [utility.syn]) are included.

```
template<size_t I, class T> struct tuple_element<I, volatile T>;  
template<size_t I, class T> struct tuple_element<I, const volatile T>;
```

- ⁵ Let *TE* denote `tuple_element_t<I, T>` of the cv-unqualified type *T*. Then specializations of each of the two templates meet the *Cpp17TransformationTrait* requirements with a member typedef `type` that names the following type:
- for the first specialization, `add_volatile_t<TE>`, and
 - for the second specialization, `add_cv_t<TE>`.
- ⁶ In addition to being available via inclusion of the `<tuple>` (22.4.2 [tuple.syn]) header, the two templates are available when any of the headers `<array>` (24.3.2 [array.syn]), `<ranges>` (ranges.syn), or `<utility>` (22.2.1 [utility.syn]) are included.

D.16 [depr.variant] Variant

- ¹ The header (22.6.2) has the following additions:

```
namespace std {  
    template<class T> struct variant_size<volatile T>;  
    template<class T> struct variant_size<const volatile T>;  
    template<size_t I, class T> struct variant_alternative<I, volatile T>;  
    template<size_t I, class T> struct variant_alternative<I, const volatile T>;  
}
```

```
template<class T> struct variant_size<volatile T>;  
template<class T> struct variant_size<const volatile T>;
```

- ² Let *VS* denote `variant_size<T>` of the cv-unqualified type *T*. Then specializations of each of the two templates meet the *Cpp17UnaryTypeTrait* requirements with a base characteristic of `integral_constant<size_t, VS::value>`.

```
template<size_t I, class T> struct variant_alternative<I, volatile T>;  
template<size_t I, class T> struct variant_alternative<I, const volatile T>;
```

- ³ Let *VA* denote `variant_alternative<I, T>` of the cv-unqualified type *T*. Then specializations of each of the two templates meet the *Cpp17TransformationTrait* requirements with a member typedef `type` that names the following type:

- for the first specialization, `add_volatile_t<VA::type>`, and
- for the second specialization, `add_cv_t<VA::type>`.

D.22.2 [depr.atomics.volatile] Volatile access

- ¹ If an atomic specialization has one of the following overloads, then that overload participates in overload resolution even if `atomic<T>::is_always_lock_free` is false:

```
void store(T desired, memory_order order = memory_order::seq_cst) volatile noexcept;
T operator=(T desired) volatile noexcept;
T load(memory_order order = memory_order::seq_cst) const volatile noexcept;
operator T() const volatile noexcept;
T exchange(T desired, memory_order order = memory_order::seq_cst) volatile noexcept;
bool compare_exchange_weak(T& expected, T desired,
                           memory_order success, memory_order failure) volatile noexcept;
bool compare_exchange_strong(T& expected, T desired,
                             memory_order success, memory_order failure) volatile noexcept;
bool compare_exchange_weak(T& expected, T desired,
                           memory_order order = memory_order::seq_cst) volatile noexcept;
bool compare_exchange_strong(T& expected, T desired,
                             memory_order order = memory_order::seq_cst) volatile noexcept;
T fetch_key(T operand, memory_order order = memory_order::seq_cst) volatile noexcept;
T operator op=(T operand) volatile noexcept;
T* fetch_key(ptrdiff_t operand, memory_order order = memory_order::seq_cst) volatile noexcept;
```

D.22.3 [depr.atomics.nonmembers] Non-member functions

```
template<class T>
void atomic_init(volatile atomic<T>* object, typename atomic<T>::value_type desired) noexcept;
template<class T>
void atomic_init(atomic<T>* object, typename atomic<T>::value_type desired) noexcept;
```

- × *Constraints:* For the volatile overload of this function, `atomic<T>::is_always_lock_free` is true.
- ¹ *Effects:* Equivalent to: `atomic_store_explicit(object, desired, memory_order::relaxed);`

10.3 Update cross-reference for stable labels for C++23

Cross-references from ISO C++ 2023

All clause and subclause labels from ISO C++ 2023 (ISO/IEC 14882:2023, *Programming Language — C++*) are present in this document, with the exceptions described below.

container.gen.reqmts *see*
 container.requirements.general

depr.arith.conv.enum *removed*
[depr.atomics.volatile](#) *removed*
 depr.codecvrt.syn *removed*
 depr.conversions *removed*
 depr.conversions.buffer *removed*
 depr.conversions.general *removed*

depr.conversions.string *removed*

// ...

depr.strstreambuf *removed*

depr.strstreambuf.cons *removed*

depr.strstreambuf.general *removed*

depr.strstreambuf.members *removed*

depr.strstreambuf.virtuals *removed*

depr.util.smartptr.shared.atomic *removed*

depr.tuple *removed*

depr.variant *removed*

depr.volatile.type *removed*

mismatch *see* alg.mismatch

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12 References

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